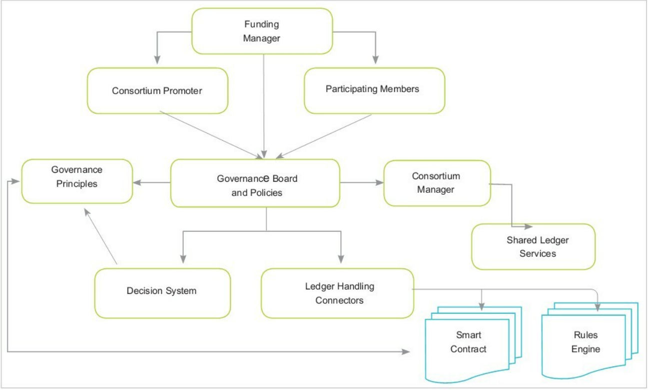

Figure 2: Blockchain consortium governance

private blockchain solutions that are to be implemented in the architecture.

Governance board and policies form the central authority to handle architecture decisions, based on the justification from the consortium promoter, participating member and founding member.

Consortium manager decides on the rules engine and the smart contract to be used in the blockchain consortium to address the key transactional requirements.

Decision system helps to choose the responsibility to be handled by different governing bodies in the blockchain consortium units.

Ledger handling connectors help to connect to the shared ledger service or cross-platform services when using a combination of public and private blockchain platforms.

Rules engine defines the set of rules to be used for transaction handling and the business logic to be used for satisfying the blockchain execution workflow.

Smart contract is defined as the proof-of-work or proof-of-stake consensus algorithm, and helps to handle the trustless entities in the blockchain platform.

The blockchain consortium governance model is designed in a way that it helps to define a hybrid blockchain platform, which is semi-decentralised, to handle the workflow under the supervision of the consortium manager. It works as a permissioned ledger for a group of consortium promoters, based on the decision system. This makes the consortium behave like a multi-party consensus platform. Every transaction and operation is verified by a special group of pre-approved nodes, and every node is not allowed to do this kind of special approval. END

References

[1] *https://www.finextra.com/blogposting/18543/blockchain-consortia-need-good-governance-but-how#:~:text=Blockchain%20consortia%20are%20defined%20as,key%20infrastructure%20and%20also%20operating*

[2] *https://academy.binance.com/en/articles/private-public-and-consortium-blockchains-whats-the-difference*

[3] *https://101blockchains.com/blockchain-consortium/*

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